

Huido Choi

M.S. Student · CLIM Lab (Global Climate Change, Innovative Monitoring & Modeling Lab)

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Portfolio <https://portfolio-eight-ruddy-87.vercel.app> Blog <https://zxsa716.tistory.com>

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EDUCATION

M.S. in Climate & Environmental Science, Kookmin University Feb 2026 – Present

Dept. of Climate Technology Convergence, Graduate School · CLIM Lab (Global Climate Change, Innovative Monitoring & Modeling Lab) · Advisor: Prof. Chul-Hee Lim

B.S. in Forestry, Environment and Systems, Kookmin University Mar 2021 – Feb 2026

Minor: School of Software Convergence · Interdisciplinary Major: Climate Big Data

GPA 4.19 / 4.5 (Major 4.3 / 4.5) · 144 credits

Boseong High School, Seoul Mar 2018 – Feb 2021

RESEARCH INTERESTS

Climate change risk assessment (IPCC AR6 framework) · Explainable AI (XAI) and graph neural networks (GAT, GCN) · Climate justice and adaptation policy evaluation · Remote sensing-based forest and urban monitoring

PUBLICATIONS

Climate justice through explainable graph neural networks: A spatiotemporal attention-based urban heat risk assessment under IPCC AR6 framework 2026

Choi, H., Park, J. S., & Lim, C.-H. (2026) · Urban Climate, Article 102981 · First author

Impact Factor 6.9, CiteScore 12.2, top 6% in field · <https://doi.org/10.1016/j.uclim.2026.102981>

RESEARCH EXPERIENCE

Graduate Researcher (M.S. Student) · CLIM Lab, Kookmin University Feb 2026 – Present

Undergraduate Researcher · CLIM Lab, Kookmin University Feb 2025 – Feb 2026

Winter Research Intern · CLIM Lab, Kookmin University Dec 2024 – Jan 2025

RESEARCH PROJECTS

Funded R&D Projects (Research Assistant / Participating Researcher)

1. Inequality and the Value of Adaptation in the Era of Climate Crisis: A Spatial Approach for Climate Justice

Ministry of Education · National Research Foundation of Korea | Research Assistant | Mar 2025 – Apr 2025

2. Development of a Digital Innovation-Based Policy Platform for Mental Health Resilience in the Era of Climate Crisis

National Research Foundation of Korea | Research Assistant | Jun 2025 – Mar 2026

3. Development of an Integrated Forest Assessment Model Based on AI and Climate Technology-Policy Convergence

Ministry of Science and ICT (Outstanding Young Researcher Program) | Research Assistant | May 2025 – Feb 2026

4. Optimal Access/Evacuation Route Analysis for Wildfire Suppression Resources and Stakeholder-Tailored Information Sharing Technology

Korea Forest Service · Korea Forestry Promotion Institute | Research Assistant | Jul 2025 – Feb 2026

5. Field-Tailored Workforce Training for the Forest Industry (R&D): Integrated Landslide Management Specialists

Korea Forest Service | Research Assistant | Jan 2026 – Dec 2026

6. Integrated Landslide Management Specialist Training Center

Korea Forest Service | Participating Researcher | Jan 2026 – Present

7. Development of AI-Based Regional Forest Road Network Planning Technology

Korea Forest Service · Korea Forestry Promotion Institute | Research Assistant | Apr 2026 – Present

8. Development of Ecosystem Restoration Technologies for Enhancing Ecosystem Value

Ministry of Climate, Energy and Environment · Korea Environmental Industry & Technology Institute | Research Assistant | Apr 2026 – Present

9. Technology for Analyzing Corporate Financial Impacts of Physical and Transition Climate Risks

Ministry of Climate, Energy and Environment · Korea Environmental Industry & Technology Institute | Research Assistant | Apr 2026 – Present

10. Kookmin University Carbon Sink Specialized Graduate School Program

Korea Forestry Promotion Institute | Participating Researcher | Jun 2026 – Present

Commissioned Research Projects

1. Translation of UNFCCC Expert Training Courses

National Institute of Forest Science | Research Assistant | Oct 2024 – Nov 2024

2. Analysis of Support Systems for Overseas Forest Resource Development

Korea Forestry Promotion Institute | Research Assistant | Oct 2025 – Dec 2025

3. Training of Bioclimatic Research Specialists (2026)

National Institute of Biological Resources | Principal Investigator | Apr 2026 – Dec 2026

4. Climate Risk Assessment under Compound Disasters (2026)

National Institute of Environmental Research | Research Assistant | Jun 2026 – Dec 2026

CONFERENCE PRESENTATIONS

Urban Heat Island Risk Assessment Using Spatiotemporal GCN

Jun 2025

Korean Society of Climate Change Research (KSCCR) 2025 Spring Conference | Poster, first author · Best Poster Paper Award

Cost and Effectiveness Evaluation of Adaptation for Mitigating Climate Inequality and Flood Risk: Focusing on Districts of Seoul

Jun 2025

KSCCR 2025 Spring Conference | Oral, second author · Best Presentation Paper Award

Eco-Community Center Development Plan Utilizing Vacant Houses in Jeongneung 3-dong

Nov 2025

2025 HUSS Environmental Consortium Conference "Climate Future" | Oral, first author

Explainable AI (XAI) for Climate Justice: GAT- and AR6 Framework-Based Spatiotemporal Heat Risk Assessment of Seoul

Dec 2025

KSCCR 2025 Fall Conference, Outstanding Graduate Student Program | Oral · Prize Winner

Resilience Analysis of Forest Ecosystems in Gangwon Province: EWS-Based Time-Series Analysis with AR(1) Indicators

Feb 2026

2026 Joint Conference of Korean Forest Science Societies | Poster, second author

Mortality Reduction Effects and Optimal Allocation of Heatwave Response Budgets in Seoul

2026

KSCCR 2026 Spring Conference | Oral, first author · Focusing on the 25 districts of Seoul

HONORS & AWARDS

Grand Prize (AI Service Creation Award) · 1st Forest Science AI Utilization Competition (NIFoS)	Jul 2026
Multi-objective forest management decision-support AI · team lead · ₩2M prize	
Excellence Award · 2026 MCEE AX Idea Competition (KEPCO KDN · KOEN)	Jul 2026
Quadruple verification of greenhouse-gas disclosure using satellite imagery · ₩3M prize	
Excellence Award (Korea National Park Service Chairman's Award) · 2025 Environmental Data Utilization & Analysis Contest	2025
Climate vulnerability prediction for Seoul using GAT-LSTM with climate scenarios · ₩2.5M prize	
Best Poster Paper Award · KSCCR 2025 Spring Conference	Jun 2025
Best Presentation Paper Award (second author) · KSCCR 2025 Spring Conference	Jun 2025
Prize, Outstanding Graduate Student Support Program · KSCCR 2025 Fall Conference	Dec 2025
Encouragement Award · Climate Change Response Business Idea Contest	
Climate adaptation platform "PCAR" financial product	
Excellence Award · Kookmin University Interdisciplinary Major Essay Contest	
Outstanding University Climate Activist · Climate Change Center U-SAVERS	Nov 2024

SCHOLARSHIPS

· Top Student Scholarship (70% of tuition) · Kookmin University	2024 – 2025
· Seonggok Scholarship · Kookmin University Graduate School	2025 – 2027

CERTIFICATIONS

· Preparing a Greenhouse Gas Inventory under the Enhanced Transparency Framework	Apr 2026
FAO eLearning Academy	
· Engineer Greenhouse Gas Management · Human Resources Development Service of Korea	Dec 2025
· Engineer Big Data Analysis · Korea Data Agency (K-DATA)	Dec 2025
· Industrial Engineer Forest · Human Resources Development Service of Korea	Sep 2025
· ADsP (Advanced Data Analytics Semi-Professional) · Korea Data Agency (K-DATA)	Mar 2025
· Computer Specialist in Spreadsheet & Database, Level 1 · KCCI	Sep 2024
· Korean History Proficiency Test, Level 1 · National Institute of Korean History	Dec 2023
· Ultra-light Vehicle (Drone) Pilot, Class 4 · Korea Transportation Safety Authority	Dec 2022

TECHNICAL SKILLS

Programming & Analysis	Python (PyTorch), R
AI / ML	GNN (GAT, GCN), LSTM, XAI (SHAP, GNNExplainer, Integrated Gradients)
Geospatial & Remote Sensing	ArcGIS Pro, Google Earth Engine, satellite image analysis
Web & Visualization	HTML/CSS/JavaScript, React, Leaflet.js, D3.js, Chart.js
Frameworks	IPCC AR6 risk assessment framework, SSP scenarios

SELECTED PROJECTS

· Multi-objective forest management decision-support AI (Grand Prize, 1st Forest Science AI Competition)	
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<https://mofom-ai.vercel.app>

- Personal research portfolio website

<https://portfolio-eight-ruddy-87.vercel.app>

- Website for Dept. of Climate Technology Convergence, Kookmin University

<https://cms.kookmin.ac.kr/climtech/index.do>

- Kookmin University's first 2024 Sustainability Report (Climate Crisis & ESG Management course)

<https://cms.kookmin.ac.kr/sustainability>

- Seoul climate change vulnerability web map

https://zxsa0716.github.io/vulnerability_seoul/seoul_vulnerability_dashboard.html

- Seoul youth single-person household safety-net map

<https://seoul-youth-platform.netlify.app>

- Community web game for reducing illegal dumping in Seongbuk-gu

<https://eco-quest-app.netlify.app>

- Technical blog on spatial analysis and climate data

<https://zxsa716.tistory.com>

EXTRACURRICULAR ACTIVITIES

President, 'Greenery' Academic Society, Kookmin University

Jan 2025 – Dec 2025

Dept. of Forestry, Environment and Systems · Seongbuk-gu Community Building Project (selected & executed) · 2025 Seoul Youth Volunteer Planning Corps (2nd cohort) · Community lecture, Seongbuk Village Community Program

U-SAVERS University Climate Activist, Climate Change Center

Mar 2024 – Nov 2024

Climate Change Young Leadership Academy (11th) · Plastic School Attack program leader · Outstanding Activist Award

'Green Narae' Youth Environmental Education Club, Mullae Youth Center

Feb 2024 – Dec 2024

Developed and delivered environmental education for youth with developmental disabilities · Main MC, 2024 Seoul Youth Environmental Festival

Sparta Coding Club 'Letanz' Supporters

Jan 2022 – Feb 2022

TRAINING

- Forest Big Data Analysis & Utilization Training (Korea Forestry Promotion Institute, 15 hrs)

Feb 2024

- Climate Change Young Leadership Academy, 11th (Climate Change Center, 32 hrs)

Mar – Nov 2024

- Python Practical Training (24 hrs)

Jul – Dec 2022

- Sparta Coding Club Web Development (40 hrs)

Jan – Feb 2022

VOLUNTEER EXPERIENCE

Total 180 hours — Mullae Youth Center (78) · Gayang 7 Community Welfare Center (64) · Climate Change Center (23) · Seoul Volunteer Center (13) · Hangang Project Headquarters (2)

MILITARY SERVICE

Sergeant, CBRN Battalion, 75th Division, Republic of Korea Army

Jul 2022 – Jan 2024

Honorably discharged after full service